

ON THE EXACT DISTRIBUTIONS OF THE SMALLEST ROOT AND THE TWO SMALLEST ROOTS OF A NON-CENTRAL WISHART MATRIX

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SUMMARY

The exact expression for the c.d.f. of the smallest root of a non-central Wishart matrix with covariance matrix $\sigma^2 I$ and a slight restriction on the sample size is derived. There is an infinite series involved in the formula. The convergence of the series is proved and the rate of convergence is discussed. The exact expression for the joint density of the smallest and second smallest roots, from which the density functions of the second smallest root and the sum of two smallest roots can be easily derived, is also given.

1. Motivation

The distribution function of the extreme roots of a Wishart matrix has been studied by many authors. Krishnaiah and Chang (1971), Pillai and Al-Ani (1969) derived the density function of the smallest root of the central Wishart matrix by means of zonal polynomials, but with different approaches. Hayakawa (1967) derived the density function for the largest root of a non-central Wishart matrix with covariance matrix I . Furthermore, Khatri (1972) and Waikar (1973) derived the c.d.f. of the extreme roots of the product of two Wishart matrices in different ways. More recently, Krishnaiah and Chattopadhyay (1975) used DeBruijn's Pfaffian method to derive the density of any root in the non-central case. Their results appear to be extremely complicated and may not be practical for computation purposes.

2. Notations and Lemma

Let us denote by $C_k(\cdot)$ the zonal polynomial of degree k corresponding to partition κ , while $(\cdot)_\kappa$, $\Gamma_m(\cdot)$ are all as defined in James (1964). Also let $a_{\kappa\tau}$, $g_{\tau\eta}^k$ be constants defined as in Constantine (1966):

$$(2.1) \quad \frac{C_\kappa(1+T)}{C_\kappa(1_m)} = \sum_{t=0}^k \sum_{\tau} a_{\kappa\tau} \frac{C_\tau(T)}{C_\tau(1_m)}$$

$$(2.2) \quad C_\tau(T) C_\eta(T) = \sum_\kappa g_{\tau\eta}^\kappa C_\kappa(T)$$

The constants $a_{\delta\tau}^\kappa$ will be defined by

$$(2.3) \quad C_\kappa(A \otimes B) = \sum_{\delta, \tau} a_{\delta\tau}^\kappa C_\delta(A) C_\tau(B)$$

where $A \otimes B$ is the direct sum of symmetric matrices A and B . Further, the constants $g_{\eta\mu\tau}^\gamma$ defined by the relation

$$(2.4) \quad C_\tau(I+T) C_\eta(T) C_\mu(T) = \sum_\star g_{\eta\mu\tau}^\gamma C_\gamma(T)$$

can be found from (2.1) and (2.2). It should be understood that the summation denoted by $\sum_\star$ in (2.4) is over the range of γ .

Lemma

Let $T = \text{diag. } (t_1, \dots, t_m)$ with $0 < t_1 < \dots < t_m < \infty$. Then

$$(2.5) \quad \int_{\infty > t_m > \dots > t_1 > 0} e^{-xtrT} |T|^a C_\kappa(T) \prod_{i>j} (t_i - t_j) dt_1 \dots dt_m \\ = \Gamma_m\left(\frac{1}{2}m\right) \cdot x^{-ma - (m/2)(m+1) - k} \Gamma_m\left(\frac{1}{2}(m+1) + a; r\right) C_\kappa(I_m) / \pi^{(1/2)m^2}$$

where κ is a partition of k , $a > -1$ and $x > 0$.

Proof

Directly from theorem 1 of Constantine (1963).

3. Distribution Theory

Let S be a non-central Wishart matrix $W(m, n, \sigma^2 I, \Omega)$. Then the joint density of the latent roots of S is given by James (1964) as

$$(3.1) \quad C e^{-(trS)/2\sigma^2} |S|^{(1/2)(n-m-1)} \prod_{i>j} (s_i - s_j) \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \frac{1}{\left(\frac{n}{2}\right)_\kappa (2\sigma^2)^k} \frac{C_\kappa(\Omega) C_\kappa(S)}{C_\kappa(I_m)}$$

where

$$C = \pi^{(1/2)m^2} e^{-tr\Omega} / [(2\sigma^2)^{(1/2)mn} \Gamma_m\left(\frac{1}{2}n\right) \Gamma_m\left(\frac{1}{2}m\right)]$$

and without loss of generality we can assume that S is diagonal,

$S \approx \text{diag.} (s_1, \dots, s_m)$ with $0 < s_1 < \dots < s_m < \infty$.

Theorem 1

If $n - m - 1$ is an even number then the c.d.f. of s_1 above is given by

$$(3.2) \quad 1 - C\star e^{-(mx/2\sigma^2)} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{n}{2}\right)_{\kappa}} \sum_{v=0}^{bm} \sum_{\eta} \frac{(-b)_{\eta} (-1)^v}{v!} \sum_{t=0}^k \sum_{\tau} \frac{a_{\kappa\tau}}{C_{\tau}(I_m)} \\ \cdot \sum_{\mu} g_{\mu}^{\tau} \Gamma_m\left(\frac{1}{2}(m+1), \mu\right) (2\sigma^2)^{t+v-k} x^{bm+k-t-v} C_{\mu}(I_m)$$

where κ, η, τ , and μ are partitions of k, v, t , and $t+v$ respectively,

$$C\star = e^{-tr\Omega} / [(2\sigma^2)^{bm} \Gamma_m\left(\frac{1}{2}n\right)]$$

and $b = \frac{1}{2}(n-m-1)$, b is an integer since $n-m-1$ is even.

Proof

From (3.1) we have

$$P(\text{all roots} > x) = \int_{\infty > s_m > \dots > s_1 > x} \int Ce^{-(trS)/2\sigma^2} |S|^{(1/2)(n-m-1)} \prod_{i>j} (s_i - s_j) \\ \cdot \sum_{k=0}^{\infty} \frac{1}{k!} \frac{C_{\kappa}(\Omega) C_{\kappa}(S)}{(2\sigma^2)^k \left(\frac{n}{2}\right)_{\kappa} C_{\kappa}(I_m)} ds_1 \dots ds_m, \quad x > 0.$$

Transforming to $t_i = \frac{s_i}{x} - 1$, with Jacobian $= x^m$, and letting $\text{diag.} (t_1, \dots, t_m) = T$, we have

$$(3.3) \quad P(\text{all roots} > x) = \int_{\infty > t_m > \dots > t_1 > 0} \int Ce^{-(mx/2\sigma^2)} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{(2\sigma^2)^k \left(\frac{n}{2}\right)_{\kappa}} \\ \cdot x^{(mn/2)+k} e^{-(trT)x/2\sigma^2} |I+T|^{(1/2)(n-m-1)} \frac{C_{\kappa}(I+T)}{C_{\kappa}(I)} \prod_{i>j} (t_i - t_j) dt_1 \dots dt_m$$

Since $n-m-1$ is even, we have the following finite sum

$$(3.4) \quad |I+T|^{(1/2)(n-m-1)} = \sum_{v=0}^{bm} \sum_{\eta} \frac{(-b)_{\eta} (-1)^v}{v!} C_{\eta}(T),$$

since $(-b)_{\eta} = 0$ if $v > bm$. Substituting (3.4) in (3.3) and using relations (2.1) and (2.2)

in (3.3), then using lemma (2.5) to integrate out $dt_1 \dots dt_m$ we get (3.2).

In order to discuss the convergence of the series (3.2), let us write down the infinite series part as:

$$(3.5) \quad \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{n}{2}\right)_{\kappa}} \sum_{t=0}^k \sum_{\tau} \frac{a_{\kappa\tau}}{C_{\tau}(I_m)} \sum_{\mu} g_{\tau\eta}^{\mu} \left(\frac{m+1}{2}\right)_{\mu} (x/2\sigma^2)^{k-t} C_{\mu}(I_m)$$

(Notice that $\Gamma\left(\frac{m+1}{2}, \mu\right) = \Gamma_m\left(\frac{m+1}{2}\right) \left(\frac{m+1}{2}\right)_{\mu}$).

Now,

$$(3.6) \quad \left(\frac{x}{2\sigma^2}\right)^{k-t} < \begin{cases} 1 & \text{if } x \leq 2\sigma^2 \\ \left(\frac{x}{2\sigma^2}\right)^k & \text{if } x > 2\sigma^2 \end{cases}$$

Denote $\max\left(1, \left(\frac{x}{2\sigma^2}\right)^k\right)$ as w_k and note that

$$(3.7) \quad \left(\frac{m+1}{2}\right)_{\mu} \leq \left(\frac{m+1}{2}\right)^{v+k}$$

From (3.6), (3.7) and (2.2) we have that (3.5) is less than or equal to

$$\begin{aligned} & \left(\frac{m+1}{2}\right)^v \sum_{k=0}^{\infty} \frac{\left(\frac{m+1}{2}\right)^k w_k}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{n}{2}\right)_{\kappa}} \sum_{t=0}^k \sum_{\tau} \frac{a_{\kappa\tau}}{C_{\tau}(I_m)} C_{\eta}(I_m) C_{\tau}(I_m) \\ &= \left(\frac{m+1}{2}\right)^v \sum_{k=0}^{\infty} \frac{\left(\frac{m+1}{2}\right)^k w_k}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{n}{2}\right)_{\kappa}} 2^k C_{\eta}(I_m) \quad (\text{by (2.1)}) \\ &\leq \begin{cases} \left(\frac{m+1}{2}\right)^v C_{\eta}(I_m) \text{EXP} \left[\frac{(m+1)x}{2\sigma^2} \text{tr} \Omega\right] & \text{if } x > 2\sigma^2 \\ \left(\frac{m+1}{2}\right)^v C_{\eta}(I_m) \text{EXP} [(m+1) \text{tr} \Omega] & \text{if } x \leq 2\sigma^2 \end{cases} \end{aligned}$$

since $\left(\frac{n}{2}\right)_{\kappa} \geq 1$. In fact, we showed that (3.5) is uniformly convergent for $x \in (0, L]$ for any L which enables us to differentiate term by term to derive the density function as the following corollary.

Corollary

Under the same assumption as in Theorem 1, the density function of s_1 is given by

$$\begin{aligned}
 (3.8) \quad & C_{\star} e^{-(ms_1)/2\sigma^2} \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{1}{2}n\right)_{\kappa}} \sum_{v=0}^{bm} \sum_{\eta} \frac{((-1/2)(n-m-1))_{\eta} (-1)^v}{v!} \sum_{t=0}^k \sum_{\tau} \frac{a_{\kappa\tau}}{C_{\tau}(I_m)} \\
 & \cdot \sum_{\mu} g_{\tau\eta}^{\mu} \Gamma_m\left(\frac{1}{2}(m+1), \mu\right) (2\sigma^2)^{t+v-k} s_1^w [(w-1)s_1^{-1} - \frac{m}{2\sigma^2}] C_{\mu}(I_m)
 \end{aligned}$$

where $w = \frac{1}{2}m(n-m-1) + k - t - v$ (notice that $w \geq 0$ since $k \geq t$ and $\frac{1}{2}m(n-m-1) \geq v$).

The q^{th} moment of s_1 can be derived easily from (3.8)

Theorem 2

Under the same assumption as in Theorem 1, the joint density function of s_1 and s_2 (second smallest root) is given by

$$\begin{aligned}
 (3.9) \quad & C_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{C_{\kappa}(I_m)} \frac{e^{-(s_1/2\sigma^2) - (m-1)s_2/2\sigma^2}}{\left(\frac{1}{2}n\right)_{\kappa}} \sum_{\delta, \tau} a_{\delta\tau}^{\kappa} C_{\delta}(S_1) \\
 & \cdot \sum_{v=0}^{bm} \sum_{\eta} \frac{(-b)_{\eta} (-1)^v}{v!} \sum_{\ell=0}^m \sum_{\mu} \frac{(-1)_{\mu} (-1)^{\ell}}{\ell!} \Sigma_{\star} g_{\eta\mu\tau}^{\gamma} (2\sigma^2)^{r-k} \Gamma_{m-2}\left(\frac{1}{2}(m-1), \gamma\right) \\
 & \cdot C_{\gamma}(I_{m-2}) s_1^{(1/2)(n-m-1)} s_2^{w'} (s_2 - s_1)^{m-1-\ell}
 \end{aligned}$$

where $w' = (1/2)(m-1)(n-m+1) - 1 + t + \ell - r$, $S_1 = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix}$, $d+t = k$; $\gamma, \delta, \tau, \eta$ and μ are the partitions of r, d, t, v , and ℓ respectively and

$$C_1 = e^{-\text{tr}\Omega} / [(2\sigma^2)^{(1/2)(m)(n-m+1)+1} \Gamma_m((1/2)n)].$$

Proof

Using the relation (2.3) in (3.1) we have the joint density of $s_1, \dots, s_m$ as

$$\begin{aligned}
 (3.10) \quad & C \sum_{k=0}^{\infty} \frac{1}{k!} \frac{C_{\kappa}(\Omega)}{(2\sigma^2)^k} \sum_{\kappa} \frac{1}{\left(\frac{1}{2}n\right)_{\kappa}} \frac{C_{\kappa}(I_m)}{C_{\kappa}(I_m)} \sum_{\delta, \tau} a_{\delta\tau}^{\kappa} C_{\delta}(S_1) C_{\tau}(S_2) \prod_{i>j} (s_i - s_j) \\
 & \cdot e^{-(s_1+s_2)/2\sigma^2} (s_1 s_2)^{(1/2)(n-m-1)} e^{-(\text{tr}S_2)/2\sigma^2} |S_2|^{(1/2)(n-m-1)}
 \end{aligned}$$

where $S_2 = \text{diag.}(s_3, s_4, \dots, s_m)$. Transform to $s_i = (t_i + 1)s_2$, $i = 3, 4, \dots, m$ with Jaco-

bian = s_2^{m-2} . Then (3.10) becomes

$$(3.11) \quad C \sum_{k=0}^{\infty} \frac{1}{k! (2\sigma^2)^k} \sum_{\kappa} \frac{C_{\kappa}(\Omega)}{\left(\frac{1}{2}\right)_{\kappa} C_{\kappa}(I_m)} \sum_{\delta, \tau} a_{\delta\tau}^{\kappa} C_{\delta}(s_1) e^{-\frac{s_1+s_2}{2\sigma^2}} (s_1 s_2)^{(1/2)(n-m-1)} \\ \cdot e^{-\frac{(m-2)s_2}{2\sigma^2}} s_2^{(1/2)(m-2)(n+2)+t(s_2-s_1)^{m-1}} e^{\frac{s_2}{2\sigma^2} \text{tr} T} |I+T|^{(1/2)(n-m-1)} \\ \cdot |T| |I+uT| C_{\tau}(I+T) \prod_{i>j=3}^m (t_i - t_j)$$

where $T = \text{diag. } (t_3, \dots, t_m)$ and $u = \frac{s_2}{s_2 - s_1}$.

Also,

$$(3.12) \quad |I+uT| = \sum_{\ell=0}^m \sum_{\mu} \frac{(-1)_{\mu} (-u)^{\ell}}{\ell!} C_{\mu}(T),$$

since $(-1)_{\mu} = 0$ if $\ell > m$. (In fact, the above expansion is equal to

$$(3.12(a)) \quad 1 + u C_{(1)}(T) + \frac{3u^2}{4} C_{(1^2)}(T) + \dots + \frac{(-1)(1^m)}{m!} C_{(1^m)}(T))$$

Substituting (3.4) and (3.12) in (3.11) then replacing $C_{\tau}(I+T)C_{\eta}(T)C_{\mu}(T)$ by $\Sigma \star g_{\eta\mu\tau}^{\gamma} C_{\gamma}(T)$ (see (2.4)), and finally using (2.5) to integrate out T , we get (3.9).

In Theorem 2, $C_{\delta}(S_1)$ can be expressed as $\sum_{u_1+u_2=d} C_{u_1 u_2} s_1^{u_1} s_2^{u_2}$, where $C_{u_1 u_2}$

is defined as in Constantine (1963). So the part which contains s_1, s_2 in (3.9) can be written as

$$e^{-a_1 s_1 - a_2 s_2} s_1^{a_3} s_2^{a_4} (s_2 - s_1)^{a_5}$$

with a_5 an integer. Thus, we can integrate out s_1 to get the density of s_2 . Also by a simple transformation we can get the density of $s_1 + s_2$.

REMARKS

1. The convergence of the series (3.9) can be shown by a similar argument as in theorem 1.
2. The density function of s_1 can also be derived from theorem 2, but theorem 1

gives us a much simpler version than that of theorem 2.

3. For computing purposes, for example in theorem 1, the speed of convergence will depend on the value of $\frac{(m+1) \times \text{tr} \Omega}{n \sigma^2}$. Obviously, it will converge very fast when $\text{tr} \Omega$ is small or the sample size is large.

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